Financial Risk and Unemployment

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Introduction

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- Firms experience a large volatility in financial risk:
  - Real interest rate (Baa)
- How much labor market volatility can be accounted for by financial shocks?
  - A search-and-matching (DMP) model with capital
  - A simple model goes a long way

Why Baa? Examples

Mechanisms DMP

Labor productivity shock $\downarrow \rightarrow$ surplus/profits $\downarrow \rightarrow v \downarrow \rightarrow u \uparrow$

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- *Vacancy cost:* vacancy cost $\uparrow \rightarrow$ vac. return $\downarrow \rightarrow v \downarrow \rightarrow u \uparrow$

Why Financial Shocks?

Summary Statistics of the US Time Series Data

	u	v	θ	r	*spread*	p
Standard Deviation	0.11	0.12	0.22	0.14	0.35	0.01
Correlation with u { two quarter lags				0.26	0.71	−0.32
Correlation with u { contemporaneous				0.32	0.62	0.05

- Quarterly moments: 1982 – 2012
- All stochastic processes in HP-log deviations

Graphs

Literature

- DMP with productivity shocks:
  - Puzzle: Shimer (2005)
  - Solutions: Hall (2005), Hall & Milgrom (2008), Hagedorn & Manovskii (2008)....
  - Fundemental surplus: Ljungqvist and Sargent (2014)
- DMP with finance:
  - Wasmer and Weil (2004)
  - Petrosky-Nadeau (2014)
  - Monacelli, Quadrini and Trigari (2011)
  - Hall (2015)
  - Chugh (2013)
  - Schoefer (2015)
  - Boeri, Garibaldi and Moen (2014)
  - Midrigan, Pastorino, and Kehoe (2014)
  - ...
- Financial shocks:
  - Christiano, Eichenbaum and Trabandt (2015)
  - Jermann and Quadrini (2012)
  - ....

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- Alternative models of debt, other untargeted moments
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- Further exploration of the data:
  - Use Compustat data
  - Study the link between $\{r, emp\}$ volatility thru ratings
  - Suggestive evidence - do not claim causality

Model – Workers

- Risk-neutral workers, $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t e_t$
- Make consumption/savings choice
  - w.r.t. risk free asset with interest rate r_f
- Resources:
  - Employed: $e = w + r_f \hat{k}$
  - Unemployed: $e = b + r_f \hat{k}$

Banks

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- Free entry pins down r given x

Firms

- Unmatched: post vacancies v at a cost $c(r)$
- Matched firms: produce y using capital k

$$y = A(1 - \alpha + \alpha k^{\rho})^{\frac{1}{\rho}}$$

- Flow profits per match: $\pi = y - w - (r + \delta)k$
  - δ is the depreciation rate
- Wages are determined by Nash bargaining

Matching

- A C.R.S. matching function $M(u, v)$: new matches
- Market tightness: $\theta = \frac{v}{u}$
  - Job-finding rate for worker: $\frac{M(u,v)}{u} = \lambda^w(\theta)$
  - Job-filling rate for firm: $\frac{M(u,v)}{v} = \lambda^f(\theta)$
- Use: $M(u, v) = \frac{uv}{(u^l + v^l)^{\frac{1}{l}}}$ (Den Haan, Ramey, and Watson)
- Separation rate σ

Value Functions – Workers

Employed worker:

$$W = w + r_f k + \beta((1-\sigma)E_{x'}W' + \sigma E_{x'}U')$$

Unemployed worker:

$$U = b + r_f k + \beta(\lambda^w E_{x'}W' + (1-\lambda^w)\, E_{x'}U')$$

Value Functions – Firms

The value of a matched firm is:

$$J = y - w - (r + \delta)k + \beta \left((1 - \sigma) E_{x'} J' + \sigma E_{x'} V' \right)$$

Vacancy posting firm:

$$V = -c(r) + \beta \left(\lambda^f E_{x'} J' + \left(1 - \lambda^f \right) E_{x'} V' \right),$$

with vacancy cost: $c(r) = (r + \delta)k + c_l$

Wages – Nash Bargaining

- Wages solve: $\max_w (W - U)^{\gamma} (J - V)^{1-\gamma}$
  - where γ is the worker's bargaining weight
- The solution is: $W - U = \gamma S; \quad J = (1 - \gamma) S$
  - where $S = (W - U) + (J - V)$

Calibration Strategy

- Time period is a week
- Exogenous shocks to the financial intermediation cost x
- Normalize $y - (r + \delta)k = 1$ when $r = \bar{r}$.
- Set some parameters a priori
- Set some parameters to match data moments

A Priori

Parameter	*Meaning*	*Value*
σ	Job Separation	0.0081
β	Discount Rate	$0.99^{\frac{1}{12}}$
c	Vacancy Cost	$c_l = .11$(HM'08)
γ	Worker Bargaining Weight	0.50
$\bar{r}$	Average Interest Rate	$(1 + 0.066)^{\frac{1}{52}} - 1$
δ	Depreciation Rate	$1 - (1 - 0.06)^{\frac{1}{52}}$
A	TFP	Normalization
α	Weight on Capital	Capital share = 1/3

Financial Shocks

Free entry condition for banks:

$$r = r_f + x$$

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Guess & verify a weekly AR(1) process that matches quarterly data on r (Real Baa):

Moment	*Parameter*	*Model*	*Data*
Frequency	Weekly	Quarterly	Quarterly
Persistence	0.995	0.798	0.799
St. Dev.	0.04	0.14	0.14

▸ Why Baa?

Calibration – Matching Moments

Endogenously calibrated parameter moments:

Parameter	*Meaning*	*Value*
l	Matching Parameter	0.41
b	Unemployment Flow Utility	0.60
ρ	Elast. Subst. K, L	-2.8

Model fit:

Moment	*Data*	*Model*
Job-finding Rate	0.139	0.139
Market Tightness	0.634	0.634
Vol. Labor Productivity	0.0095	0.0095

Results – Data versus Model

		u	v	θ	y	r
St Dev	Data	0.11	0.12	0.22	0.01	0.14

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St Dev	Data	0.11	0.12	0.22	0.01	0.14
	Model	0.07	0.08	0.14	0.01	0.14
Persistence	Data	0.94	0.91	0.93	0.81	0.80
	Model	0.86	0.63	0.79	0.83	0.80

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Persistence	Data	0.94	0.91	0.93	0.81	0.80
	Model	0.86	0.63	0.79	0.83	0.80
Corr u	Data	1	-0.89	-0.97	-0.61	0.32
	Model	1	-0.76	-0.93	-0.97	0.87
Corr v	Data	-	1	0.97	0.71	-0.26
	Model	-	1	0.94	0.89	-0.88
Corr θ	Data	-	-	1	0.68	-0.30
	Model	-	-	1	0.99	-0.93
Corr y	Data	-	-	-	1	-0.16
	Model	-	-	-	1	-0.91

Table: Quarterly moments: data: 1982- 2012 versus Model

Bank Debt

Breakdown of Mechanisms

Interest rate rises:

- *Profits:* capital costs $\uparrow \rightarrow$ profits $\downarrow \rightarrow v \downarrow \rightarrow u \uparrow$
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Mechanisms	u	v	v/u
Data	0.11	0.12	0.22
All	0.07	0.08	0.14
Profit	0.05	0.06	0.10
Vacancy	0.02	0.02	0.04

Robustness

Exercise	*Robustness*	*Why*	u	v	θ	r
Data			0.11	0.12	0.22	0.14
Benchmark			0.07	0.08	0.14	0.14

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Data			0.11	0.12	0.22	0.14
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$\uparrow \delta$	δ=0.08	Robust	0.06	0.07	0.12	0.14
$\uparrow \gamma$	γ=0.72	Shimer	0.07	0.09	0.14	0.14
$\downarrow \gamma$	γ=0.28	- Shimer	0.06	0.08	0.14	0.14

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$\uparrow \gamma$	γ=0.72	Shimer	0.07	0.09	0.14	0.14
$\downarrow \gamma$	γ=0.28	- Shimer	0.06	0.08	0.14	0.14
Levels	No Log	Gali	0.07	0.08	0.14	0.9%

Check – Labor Productivity

Now replace with labor productivity shocks.

Robustness	u	v	θ	r	p
Data	0.11	0.12	0.22	0.14	0.01
Benchmark	0.07	0.08	0.14	0.14	0.01
Just p shock	0.01	0.01	0.02	0.00	0.01

Debt: Interpretation

Benchmark: Bond debt. Interest rate change during match.

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Quantitatively: Similar. Maintain same calibration.

Equations

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	Model	-	-	-	1	-0.78

Table: Quarterly moments: data: 1982- 2012 versus Model

Bond Debt

Other Untargeted Moments

interest/sales=
(int. payment + prop. inc.)/(operating surplus + dep)

Moment	*Data*	*Bond Debt*	*Bank Debt*
Average Interest/Sales	0.406	0.480	0.429
Volatility of Interest/Sales	0.013	0.034	0.019
Volatility of Labor Share	0.007	0.017	0.014

Bond debt volatility interest/sales, recalculated: 0.011

More on the Labor Share

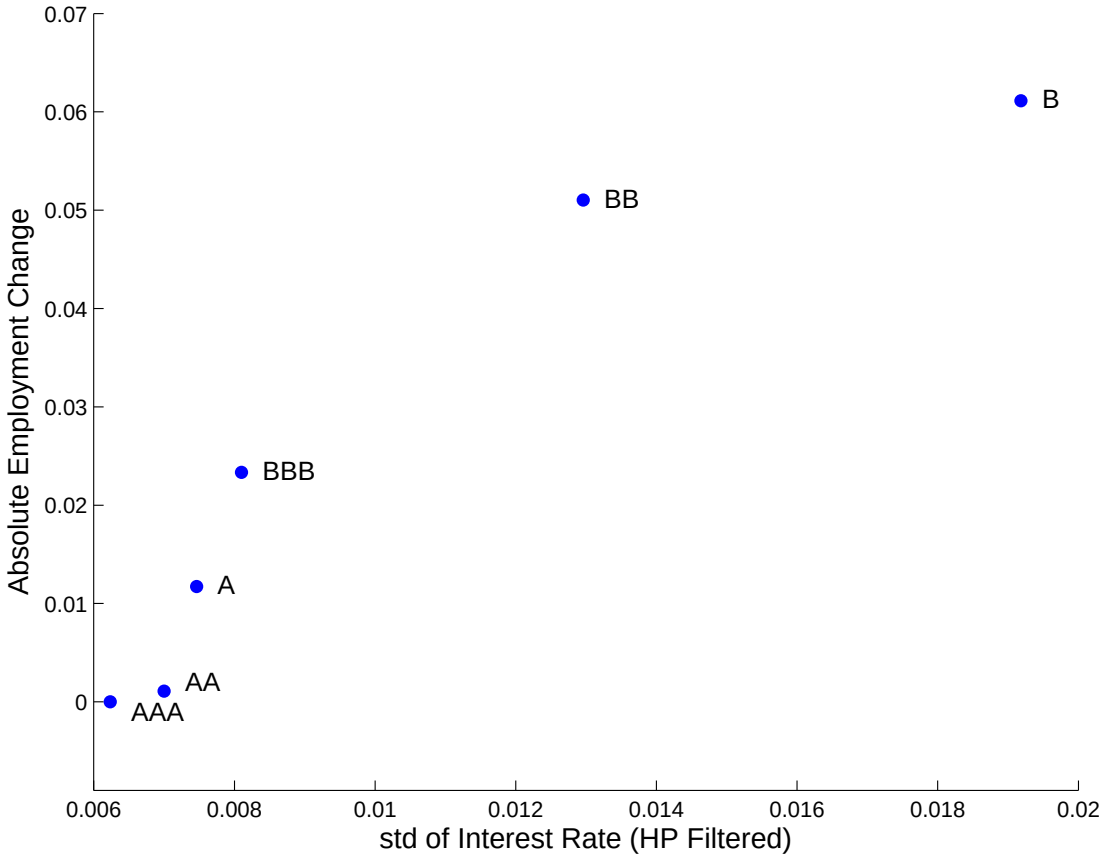
Wages determined by:

- Flow surplus: $y - (r + \delta)k - b$
- Outside option: Hall-Milgrom (2008)

Model	u	v	v/u	$\frac{w}{y}$
Data	0.11	0.12	0.22	0.007
Bond Debt: Nash Bargaining	0.07	0.08	0.14	0.017
Bank Debt: Nash Bargaining	0.07	0.08	0.14	0.014
Bank Debt: Alternating Offer	0.14	0.21	0.34	0.009

Equations

Relationship Between Interest Rate and Employment Volatility



Data

- Hypothesis A: Lower rating firms $\Rightarrow$ r volatility $\uparrow$
- Source: BofA Merrill Lynch quarterly rating-level data:
  - Credit Rating
  - Effective yield of US Corporate ratings indexes
  - 1997Q1–2015Q3

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- Hypothesis B: Lower rating firms $\Rightarrow$ Emp volatility $\uparrow$
- Source: Compustat data firm-level annual data:
  - Credit Rating
  - Employment
  - 1986–2012

Compustat - Summary Statistics: 1986-2012

Rating	Mean(Employment) (Thousands)	Std(Employment) (Thousands)	Observations All years	Observations 2008
AAA	88	100	659	10
AA	54	140	2975	72
A	34	59	8809	287
BBB	23	46	10484	504
BB	14	27	7252	351
B	9	19	6283	294
CCC	6	17	549	18
Total			37011	1536

Examples

A: Lower rating firms $\Rightarrow$ r volatility $\uparrow$

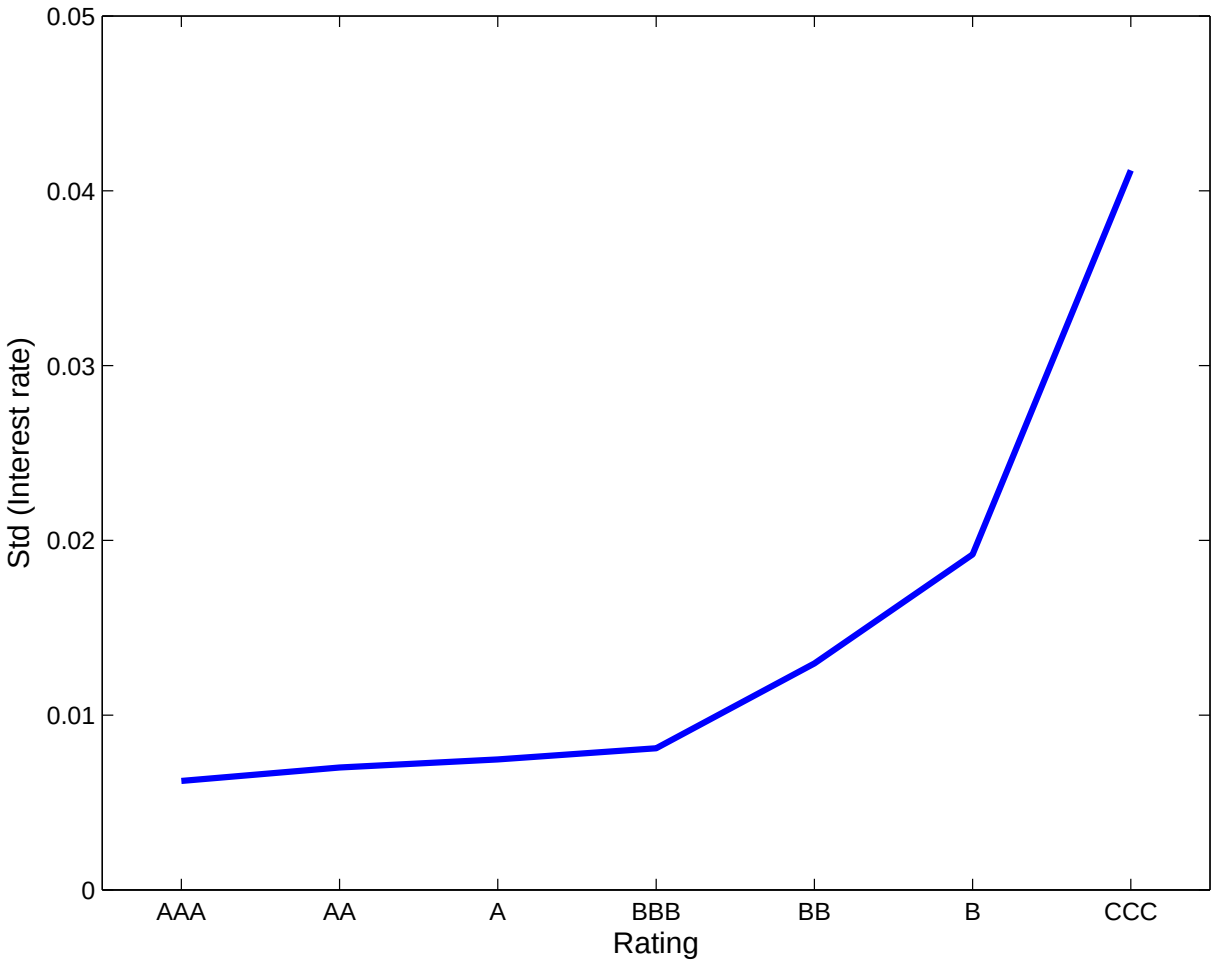

Figure: Standard deviation of (HP filtered) effective yields by rating

B: Lower rating firms $\Rightarrow$ Emp vol. $\uparrow$

- Estimate the following model:

$$y_{it} = \beta_0 + \sum_{r=1}^{7} \beta^r \text{RATING}_{it} + \sum_{t=1986}^{2012} \beta^t \text{YEAR}_t + \beta^c \text{CONTROLS}_{it},$$

  - y_{it} is the absolute % employment change for following year
  - RATING_{it} is a dummy for rating of firm i at time t (S&P)
  - YEAR_t is a dummy for each year
  - CONTROLS_{it} is a cubic polynomial in employment
  - Clustered by firm

Ratings and Employment change

$$y_{it} = \beta_0 + \sum_{r=1}^{7} \beta^r \text{RATING}_{it} + \sum_{t=1986}^{2012} \beta^t \text{YEAR}_t + \beta^c \text{CONTROLS}_{it},$$

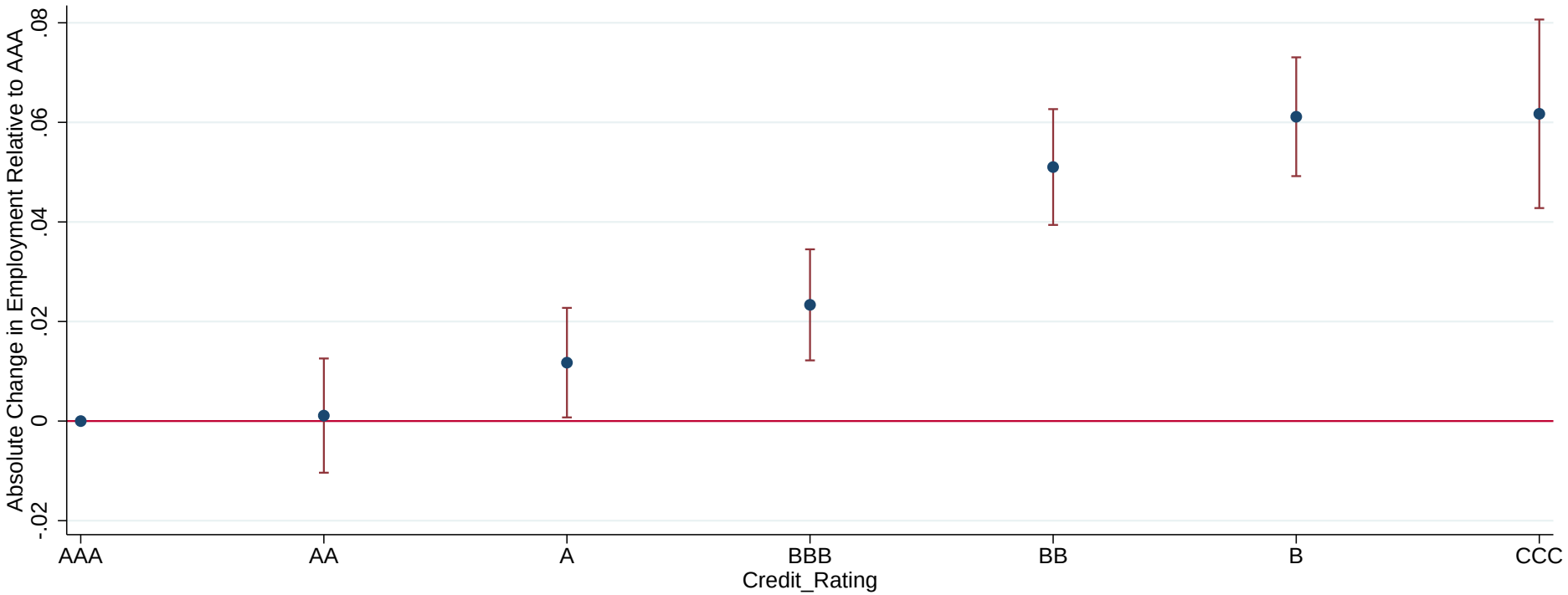

Figure: Absolute Percentage Change in Employment by Rating

Trends? An alternative measure 2008

Interest rates and Employment

Taking together the two hypotheses we arrive at:

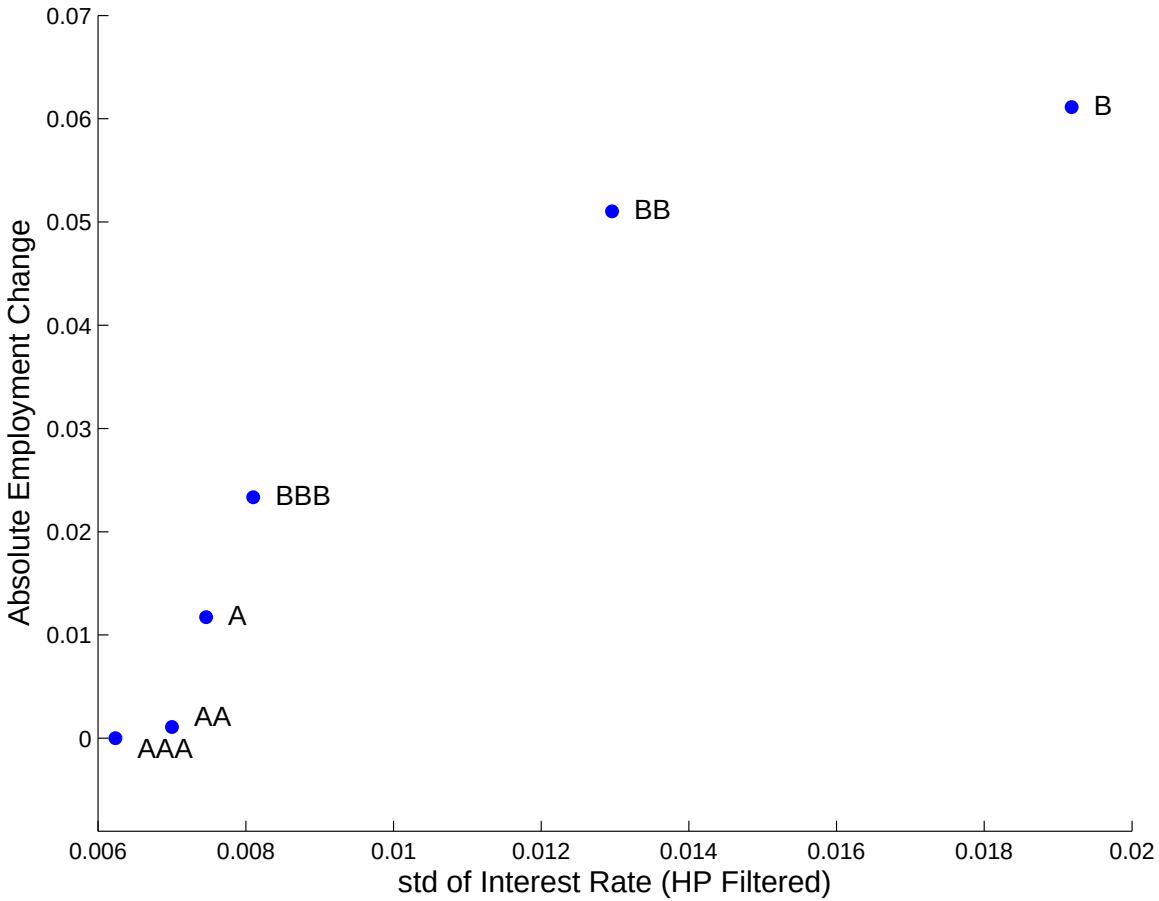

Figure: Interest rates and Employment

Conclusion

We studied:

- Mechanisms for financial risk affecting unemployment
- Quantitative effect of those shocks using DMP literature

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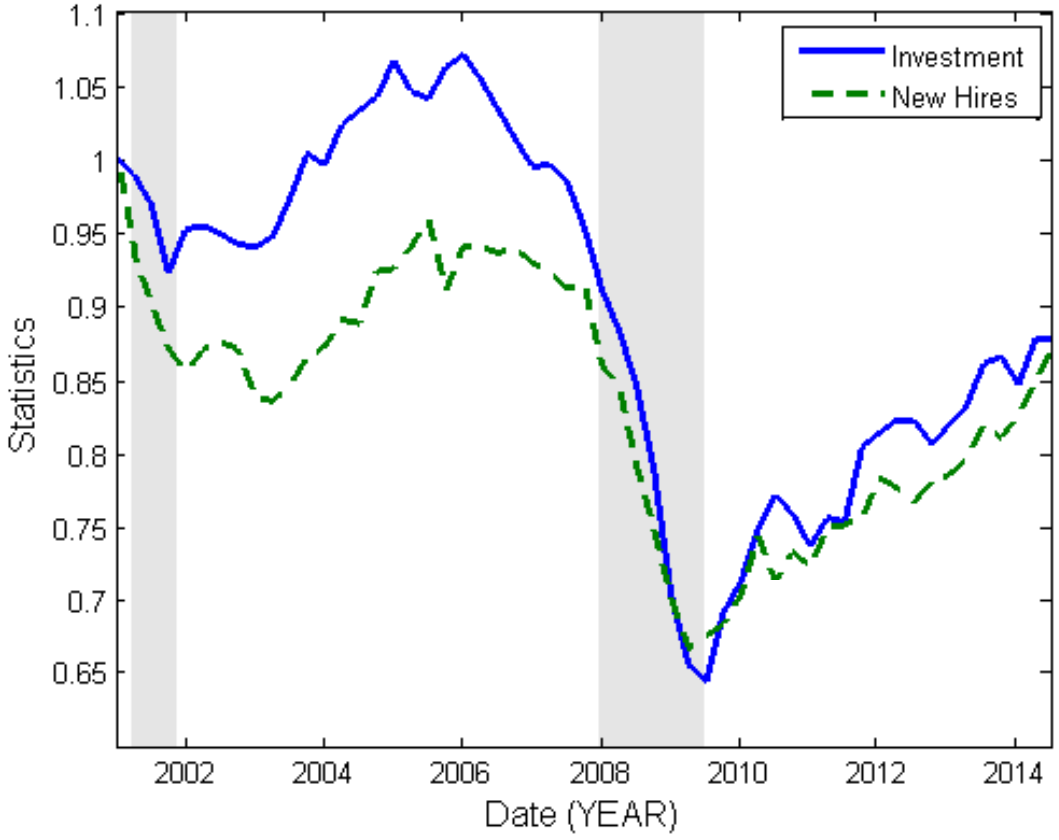
We studied:

- Mechanisms for financial risk affecting unemployment
- Quantitative effect of those shocks using DMP literature

We found:

- Financial conditions (interest rates) matter a lot
- Can think about various forms of debt
- Empirical evidence consistent

Hires and Investment



- Hires from JOLTS, Inv. is real gross private domestic
- Correlation = 0.73
- St dev of log is 0.11 for investment, 0.10 for hires

Back to Capital

Bank Debt: Equations

$$J_{x,\hat{x}} = y - w_{\tilde{x}} - (r_{\tilde{x}} + \delta)k + \beta\,(1-\sigma)\,E_{x,\hat{x}} J_{x',\tilde{x}},$$

$\tilde{x} = \min x, \hat{x}$

Back

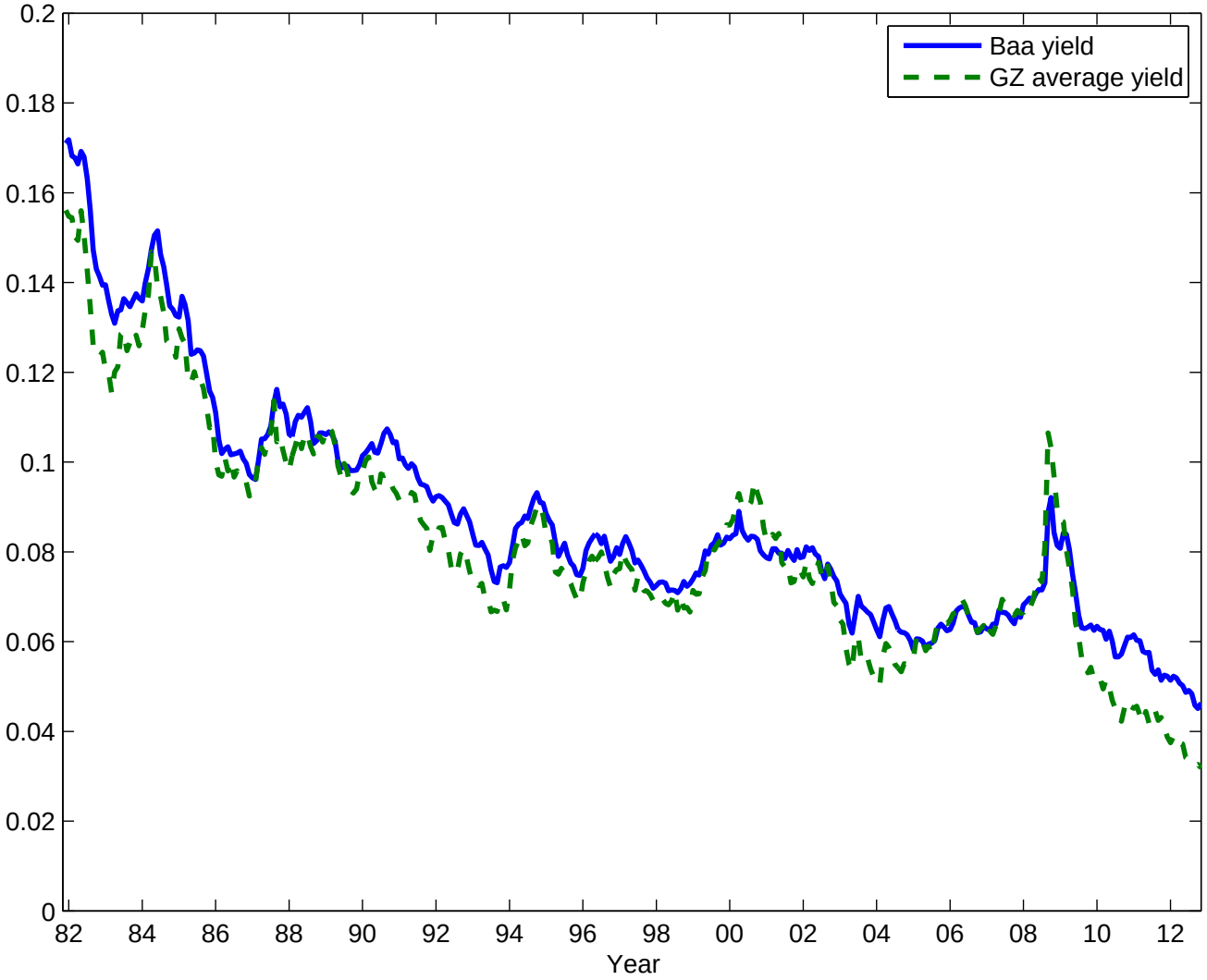
Alternating Wage Offer: Equations

$$W^{w}_{x,\hat{x}} = \nu U_x + (1-\nu)\left[b + r_f a_x + \beta\left\{(1-\sigma)\, E_{x'} W^{w^*}_{x',\tilde{x}} + \sigma E_{x'} U_{x'}\right\}\right]$$

$$J^{w^*}_{x,\hat{x}} = (1-\nu)\left[-\omega + \beta\,(1-\sigma)\, E_{x'} J^{w}_{x',\tilde{x}}\right]$$

Back

Why Baa?



Correlation: 0.97

Standard Deviation: Baa is 2.7% Gz is 2.6%

Back to intro Back to FS

Employment and Ratings - Trends?

$$y_{it} = \beta_0 + \sum_{r=1}^{7} \beta^r \text{RATING}_{it} + \sum_{t=1986}^{2012} \beta^t \text{YEAR}_t + \text{CONTROLS}_{it},$$

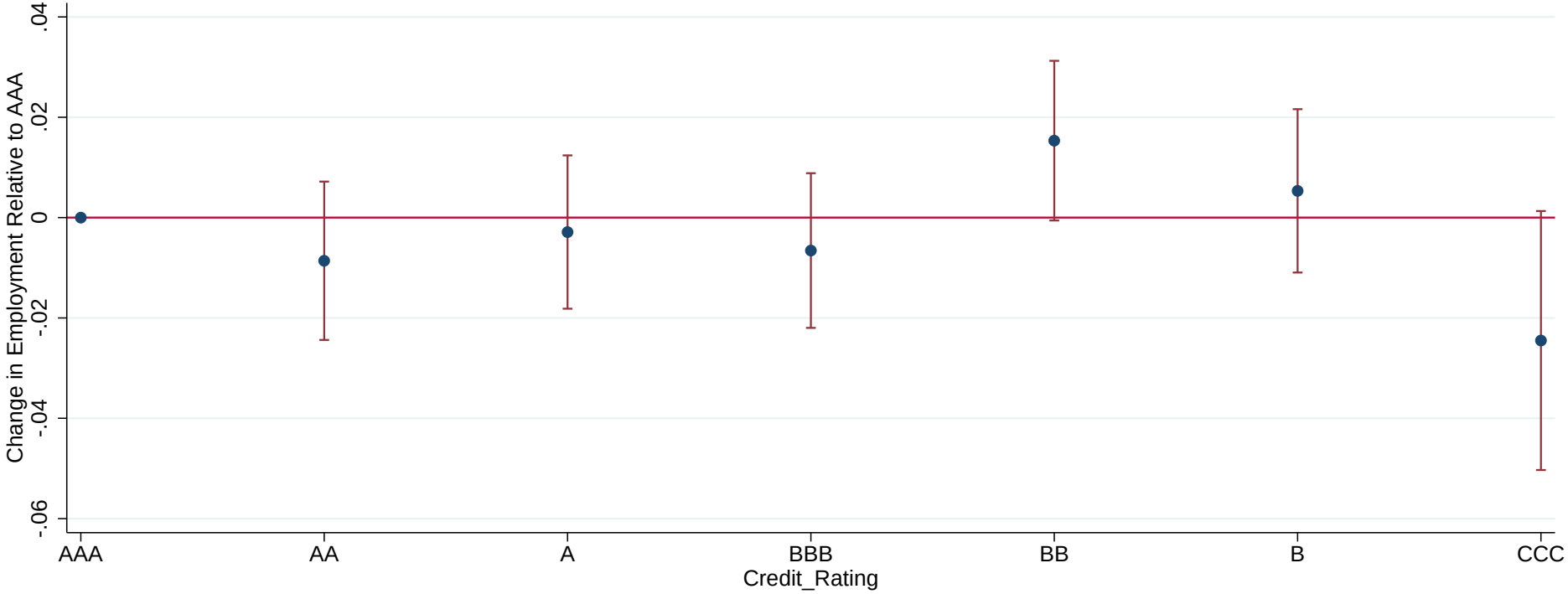


Figure: Percentage Change in Employment by Rating

Ratings and Employment change

An Alternative Measure

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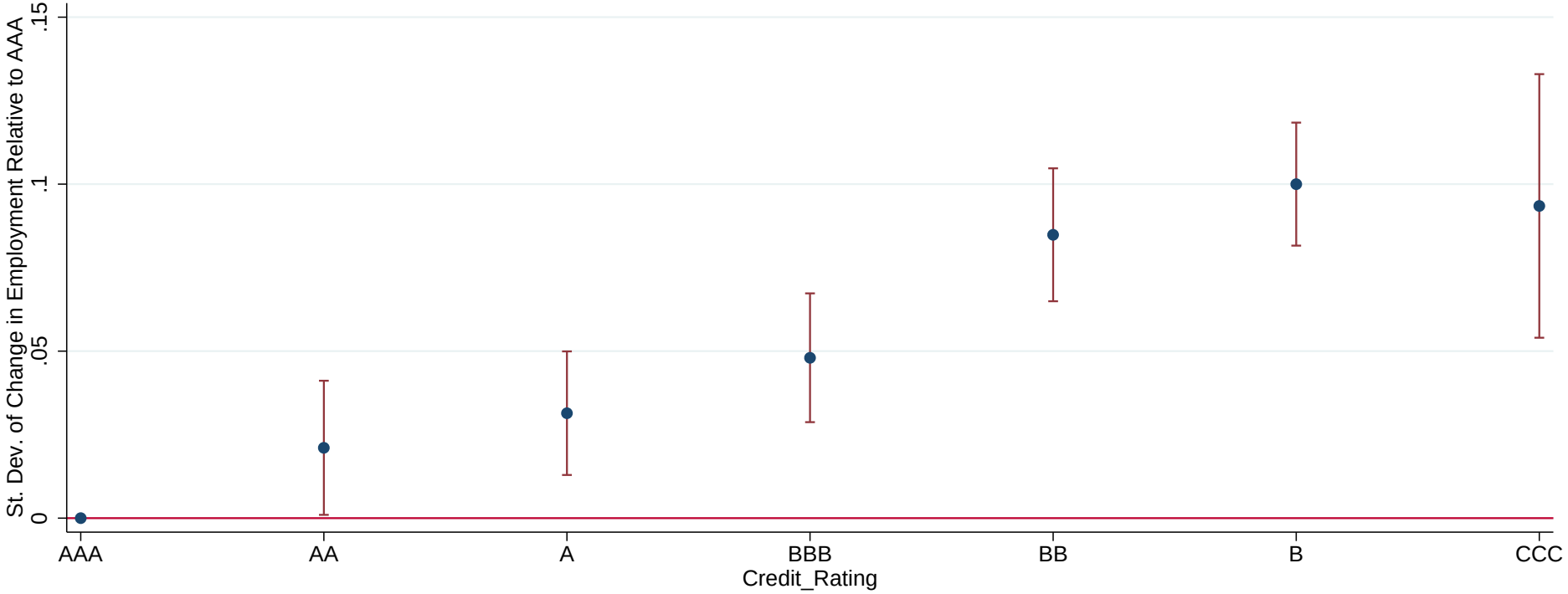

Figure: Standard deviation of Employment by Rating

The Great Recession

$$y_i = \beta_0 + \sum_{r=1}^{16} \beta^r \text{RATING}_i + \text{CONTROLS}_i,$$

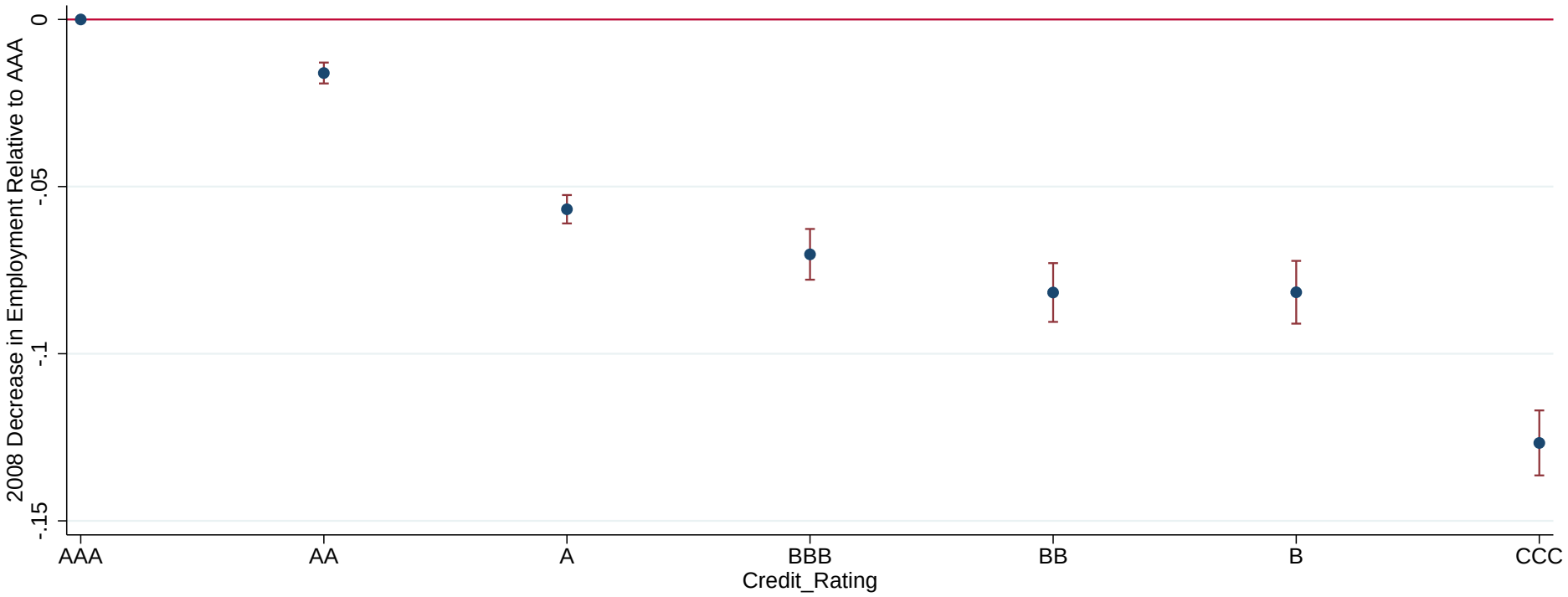

Figure: Percentage Change in Employment by Rating 2008–2009

Some Examples for Firms' Ratings

Rating	Firms
AAA	Exxon Mobil Corp, Johnson & Johnson
	Microsoft Corp
AA	Chevron Corp, Canon Inc
	Abbott Laboratories
A	Boeing Co, Campbell Soup Co
	Citigroup Inc
BBB	Block (H&R) Inc, Coca-Cola Bottling
	FedEx Corp.
BB	Macy's Inc, Sprint Corp
	Rent-A-Center Inc
B	Delta Air Lines Inc., Toys 'R' Us Inc
	Office Depot Inc
CCC	Brookstone Inc, Rotech Healthcare Inc
	E*TRADE Financial Corporation

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Unemployment, Spread, Productivity

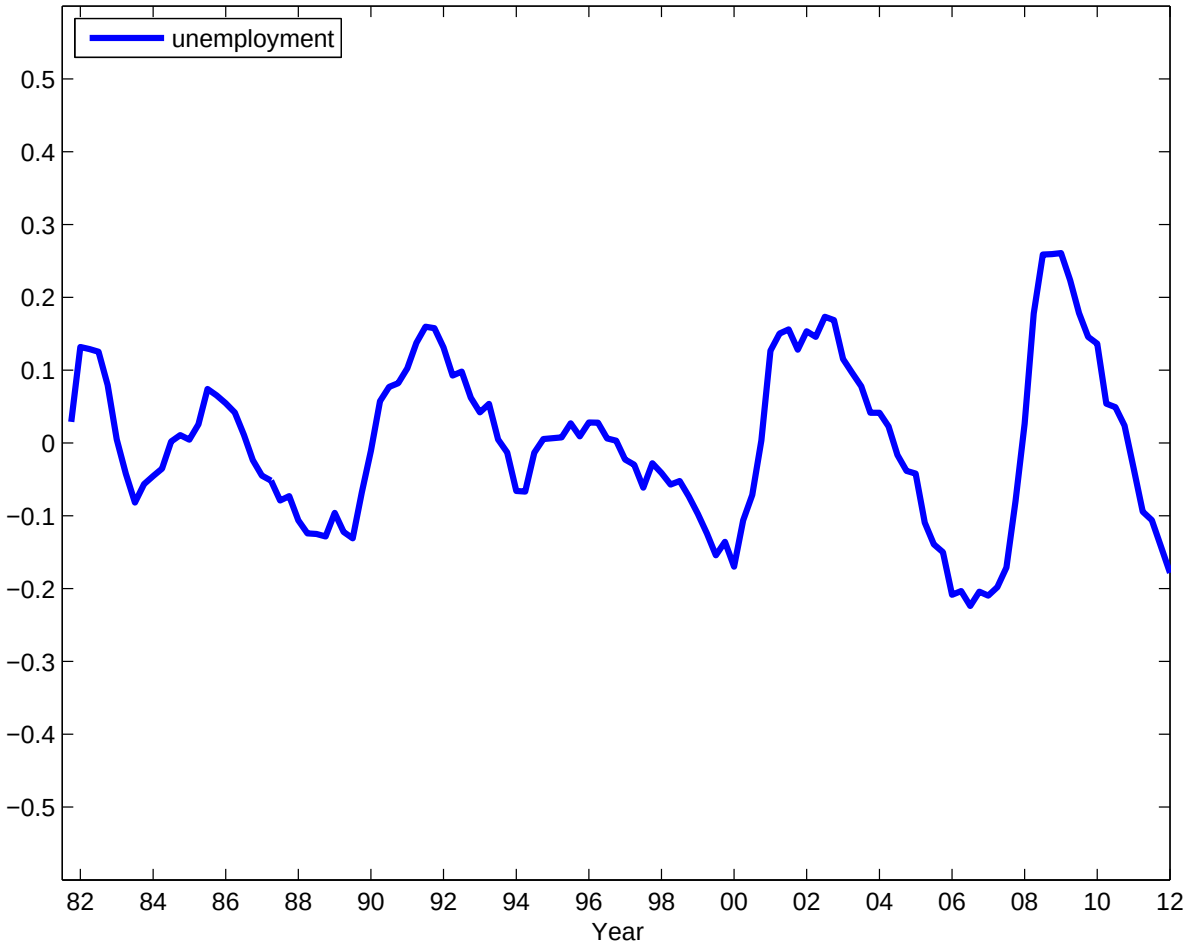

Figure: US time-series data 1982-2012
Spread and productivity are lagged by 2 quarters

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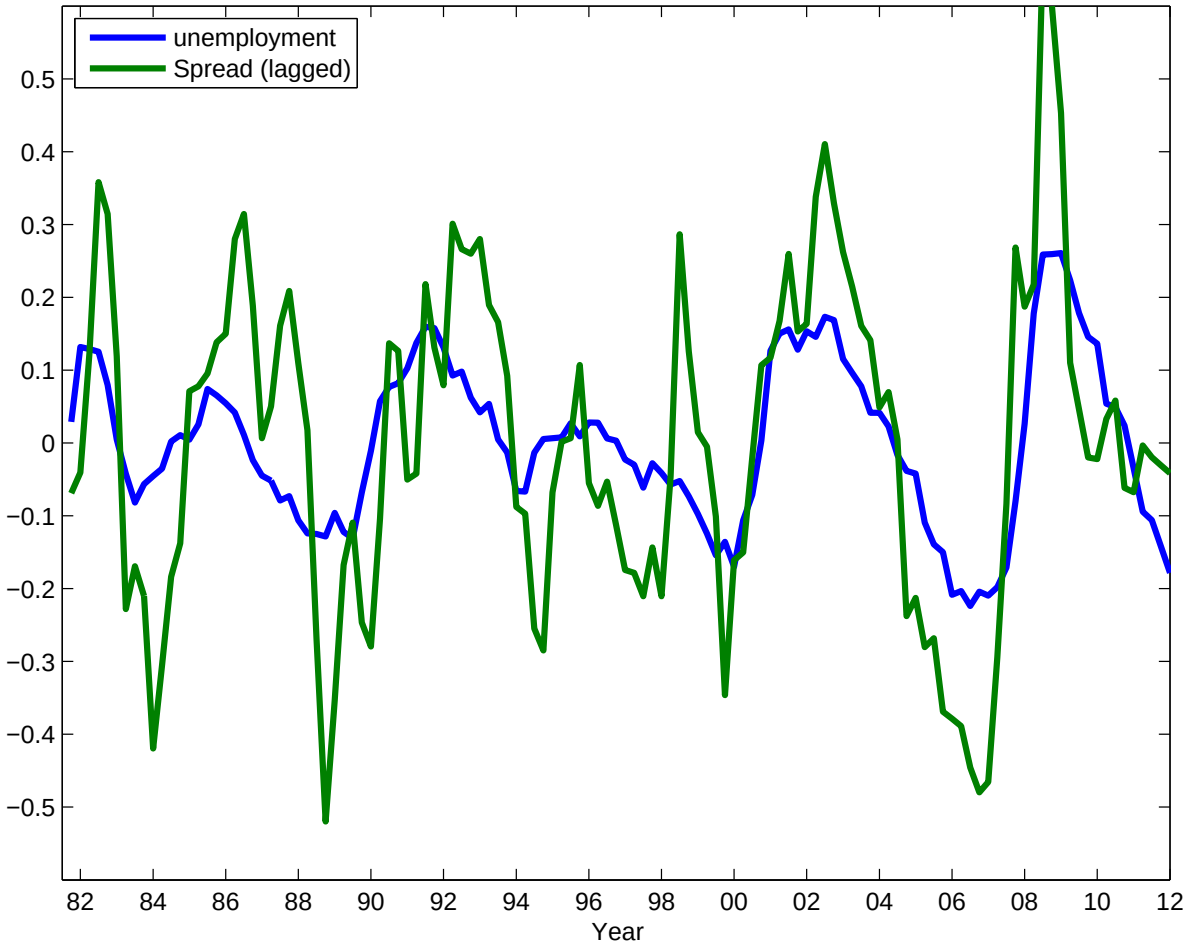
Unemployment, Spread, Productivity



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Unemployment, Spread, Productivity

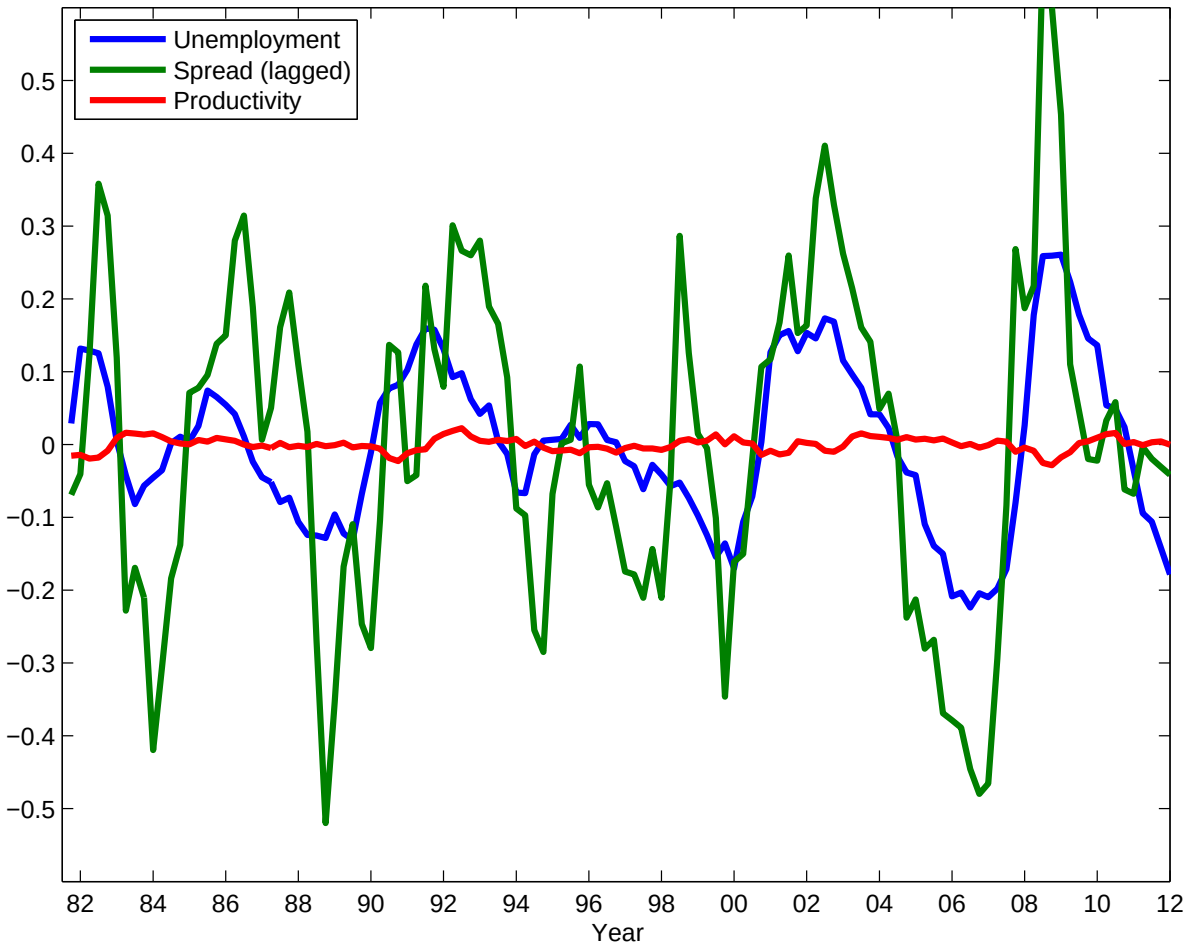

Figure: US time-series data 1982-2012
Spread and productivity are lagged by 2 quarters

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Unemployment, Interest, Productivity

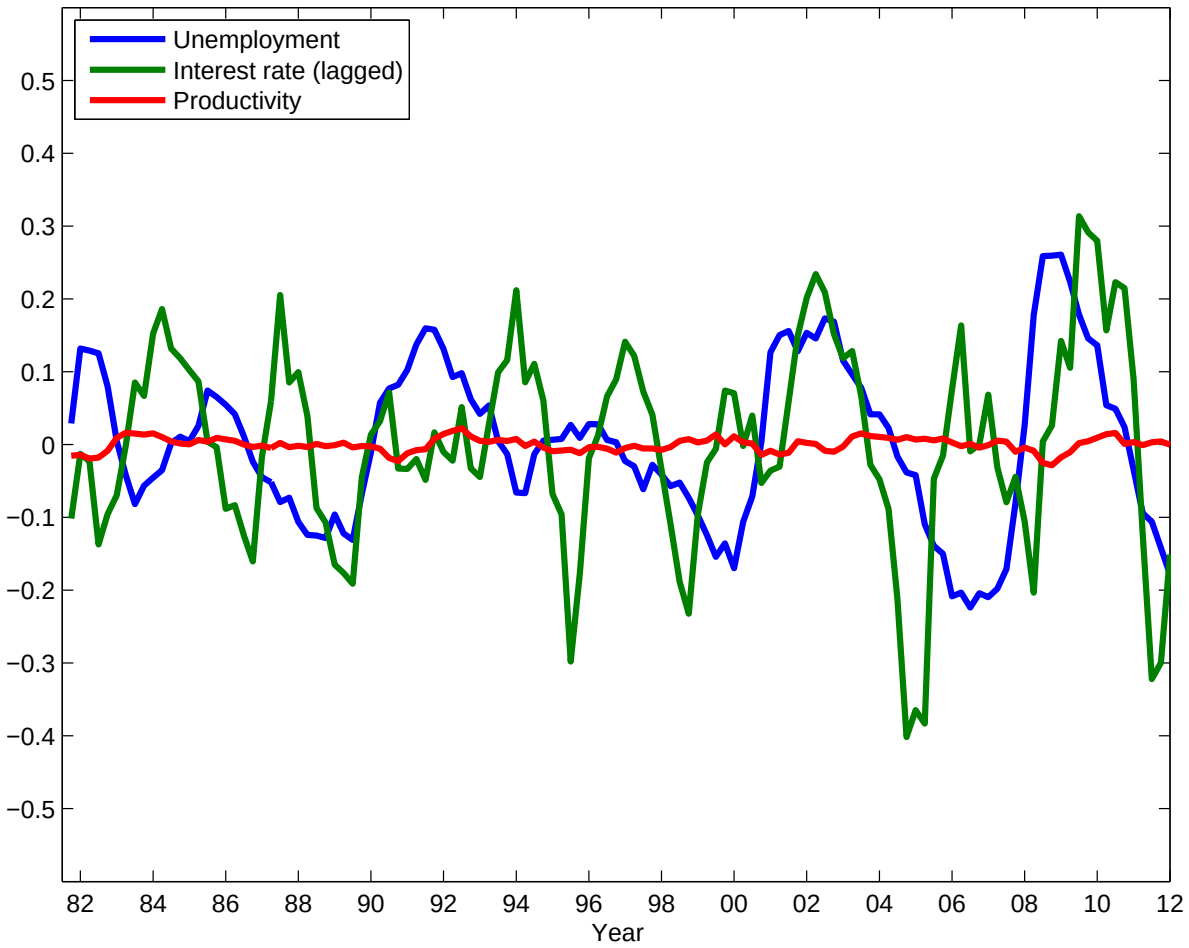

Figure: US time-series data 1982-2012
Interest rate and productivity are lagged by 2 quarters

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